MILITARY SPECIFICATION

WEBBING, TEXTILE: KNITTED, NYLON; ELASTIC; 4-1/4 INCH WIDTH

1. SCOPE

*1.1 This specification covers requirements for one type of knitted nylon, elastic webbing.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-43334	- Webbing And Tape, Textile, Packaging And Packing Of.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.
FED-STD-601	- Rubber; Sampling And Testing.

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber

Products Identification Act

(Copies may be obtained from the Federal Trade Commission, Washington, D. C. 20580,

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3. REQUIREMENTS

*3.1 Sample. Samples are furnished solely for guidance and information to the supplier (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

*3.2 First article. Unless otherwise specified (see 6.2), before production is commenced, finished webbing shall be submitted, or made available, to the contracting officer or his authorized representative for approval. The approval of the first article authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all provisions of this specification. The first article shall be manufactured by the supplier at the same facilities to be used for the manufacture of the production item.

***3.3 Material. The webbing shall be made of 50 gage extruded natural round rubber elastic strands wrapped with 70 denier and 200 denier nylon yarns. The knitting yarns shall be 210 denier nylon yarn (see Section 4, Table III for required test).**

*3.4 Design. The webbing shall conform to the requirements specified in Table I when tested as specified in Section 4, Table V.

Table I. Physical requirements

Characteristic	Requirement
Width-inches:	
-web portion	3-1/2 ± 1/16
-selvage	3/8 ± 1/32
-overall	4-1/4 ± 1/8
Thickness-inch	0.038 ± .003
Weight-ounces per linear yard	2.2 - 2.6
Total sewing sections (warp direction)	6 minimum
Number of ends per sewing section division	6 minimum
Wales per inch	22 ± 2
Ribs per inch (filling direction)	19 ± 2
Color (unless otherwise specified)	OG-106
Colorfastness to:	
-wet(dry) cleaning	Good
-crocking	Good
-light	Good
-perspiration	Good

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*3.4.1 Ultimate elongation and rupture of rubber strands. There shall be no evidence of rupture of rubber strands when the webbing is subjected to an ultimate elongation of 200 percent in the filling direction.

*3.5 Construction. The knitted webbing shall have a transverse web of interlocking and interconnecting warp and filling yarns with a selvage at each edge. The laid in elastic warps shall consist of a rubber core wrapped first with 2-ply, 70 denier, 1/2 twist nylon and then wrapped with single ply, 200 denier, 1/2 twist nylon (see Section 4, Table III for required test).

*3.6 Length of roll. Unless otherwise specified (see 6.2), the webbing shall be furnished in rolls of not less than 33 yards. There shall not be more than 3 pieces per roll and the minimum length of any one piece shall be 10 yards.

*3.7 Identification tickets. Each roll of webbing shall have an identification ticket attached to the roll in accordance with MIL-W-43334.

*3.8 Fiber identification. Each roll of webbing shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

*3.9 Workmanship. The finished elastic webbing shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable quality levels (AQL).

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

*4.2 Inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

*4.2.1 First article inspection. The first article submitted in accordance with 3.2 shall be visually inspected for appearance and color. The first article shall be tested for physical and chemical properties in Section 3 in accordance with methods specified in 4.3.

*4.2.2 Testing of components. Components listed in Table II shall be tested for the characteristics specified herein in accordance with the referenced test methods of FED-STD-191. The sample unit shall be one cone, tube, or spool and the lot size shall be expressed in units of one cone, tube, or spool each. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size shall be as follows:

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Lot size	Sample size
800 or less	2
801 up to including 22,000	3
22,001 and over	5

Table II - Testing of components

Characteristic	Req. para.	Test, method
Nylon yarn:		
Material Identification	3.3	1/
Denier	3.3	4021
Turns, twist	3.5	4052 2/
Rubber Yarn:		
Material identification	3.3	1/
Gage	3.3	4.3.1

1/ The supplier shall submit a certificate of compliance for the characteristic listed. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

2/ One determination per sample unit and the results reported as "pass or fail".

*4.2.3 Examination of the end item. The webbing shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.2.3.1, 4.2.3.2 and 4.2.3.3.

*4.2.3.1 Yard-by-yard examination. The yardage specified in Table III shall be inspected on one side and visual defects classified as follows. The side examined shall be alternated on every other roll. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the webbing, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each wale-wise (lengthwise) yard or fraction in which it occurs. The sample unit for this examination shall be one linear yard. The AQL for this examination shall be 1.0 major defects and 4.0 total defects (major and minor combined) per 100 units (yards). The sample size shall be based upon inspection level II, MIL-STD-105. The sample size shall be expressed in units of 1 yard. The number of rolls from which the sample is to be selected shall be in accordance with Table III. An approximately equal number of yards shall be examined in each roll in the sample.

Defects	Classification Major	Minor
Any hole, cut, tear	X	
Run or dropped stitch, end cut	X	
Tuck stitch, pulled stitch, or needle streak		X
Abrasion mark, resulting in a weak area	X	
Thin or uneven fabric which may develop into a hole	X	
Coarse or fine yarn 1/	X	
Tight twist yarn		X
Slubs or slugs or gouts, more than twice the size of normal yarn		X

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Defects	Classification Major	Minor
Foreign matter 1/		X
Knots, oversize, extending above the surface of cloth		X
Kinks or untrimmed ends		X
Crease or wrinkle embedded		X
Stretch (stitch tension) knitting tension too tight or too loose	X	
Shade bar 1/		X
Edges cut or frayed	X	
Spot or stain 1/	X	

1/ Clearly visible at normal inspection distance (approximately 3 feet).

Table III - Sample sizes

Lot Size in yards	Sample size	Max. No. of defects acceptable in Spl 1/
Up to 1200 2/	3	0
1201 up to & including 3200	5	0
3201 up to & including 10,000	8	0
10,001 up to & including 35,000	13	0
35,001 up to & including 150,000	20	1
150,001 and over	32	2

1/ Except that the acceptance number shall be zero for color and uniformity of shade defects found in the overall examination (4.2.3.2).

2/ If a lot contains fewer than 3 rolls or spools, each roll or spool in the lot shall be examined.

*4.2.3.2 Overall examination. Each defect listed below shall be counted no more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size (number of rolls selected as sample) for this examination and the number of defects acceptable shall be as shown in Table III.

Width less than specified.
Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the piece.
Off shade, shaded end to end, side to side, side to center or throughout the roll.
Mottled, streaky or cloudy appearance.
Edges cut, frayed, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll.

*4.2.3.3 Examination of length. The roll shall be examined for gross length. Any gross length found to be less than the minimum specified herein or more than two yards below the gross length marked on the ticket shall be considered a defect with respect to length. The sample unit for this examination shall be one roll. The sample size (number of rolls selected as sample) for this examination and the acceptance number shall be as shown in Table III. The lot shall be unacceptable if the total of the actual gross length of rolls in the sample is less than the total of the gross lengths marked on the roll tickets.

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*4.2.4 Examination of preparation for delivery requirements. An examination shall be made to determine compliance with packaging, packing and marking requirements of Section 5 of this specification.

*4.3 Testing of the end item. The methods of testing specified in Table IV shall be in accordance with FED-STD-191. When the data in the "results reported as" column are not specified in the table they shall be as required by the referenced test method. The physical and chemical values specified in Section 3 apply to the result(s) of the determination(s) made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 3-1/2 yards of finished webbing. The lot size shall be expressed in units of 1 linear yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified herein (see 4.2.2 for sample size).

Table IV - Testing of the end item

Characteristic	Req. para.	Test method	Results reported as
Weight, oz/lin yd	3.4	5040	
Total sewing sections (warp)	3.4	Visual	Pass or fail
Number of ends per sewing section div.		Visual	Pass or fail
Waler per inch	3.4	5070	
Ribs per inch (filling)	3.4	5070	
Colorfastness to:			
Wet (dry) cleaning	3.4	5622	
Crocking	3.4	5651	
Light	3.4	5660 1/	
Perspiration	3.4	5680	
Ultimate elongation and rupture of rubber strands	3.4.1	4.3.2	Pass or fail

1/ Except that the specimen shall be exposed for 40 standard fading hours.

*4.3.1 Gage of rubber. The gage of the rubber shall be determined by counting the actual number of strands, laid side by side, contained in 1 inch. The gage is equivalent to the actual number of the rubber strands contained in 1 inch.

*4.3.2 Ultimate elongation test. The test for ultimate elongation shall be in accordance with Method 4121 of FED-STD-601. Determination of the rupture of the rubber strands shall be made during the test for ultimate elongation.

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*5. PREPARATION FOR DELIVERY

*5.1 Packaging and packing. Packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-W-43334 (see 6.2).

***5.2 Marking. In addition to any special markings required by the contract or order, shipments shall be marked in accordance with MIL-W-43334. The item identification nomenclature to be marked on rolls and shipping containers shall be: Webbing, Textile: Knitted, Nylon, Elastic, 4-1/4 inch width, OG-106.**

6. NOTES

***6.1 Intended use. The webbing is intended for use on Marine Corps body armor lower torso. Specifically, the webbing is used for joining the left and side back part of lower torso of the body armor.**

*6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

***6.2.1 Procurement requirements:**

(a) Title, number and date of this specification.
(b) Whether first article is required (see 3.2).
(c) Length of roll desired (see 3.6) if other than specified.
(d) Selection of applicable levels of packaging and packing (see 5.1).

***6.2.2 Contract data requirements. Data conforming to Data Item Descriptions R-01, R-02, R-04, T-01, T-02, T-03 and T-04 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.**

***6.3 Sample and shade sample. For information regarding the availability of sample webbing, and of the standard shades specified, address inquiry to the procuring activity issuing the invitation for bids.**

***6.4 First article. Examinations, tests and approval shall be as specified by the contracting officer (see 3.2).**

***6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience *only and the Government assumes no liability whatsoever* for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship of the last previous issue.**

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